

A Practical Guide for Spotting Spider Mites

Life Cycle • Adult Morphology • Leaf Damage

- Two-Spotted Spider Mite (TSSM, *Tetranychus urticae*)
- Four-Spotted Spider Mite (FSSM, *Tetranychus turkestani*)
- New Red Spider Mite (new red *Tetranychus* sp.)

Grbić Lab

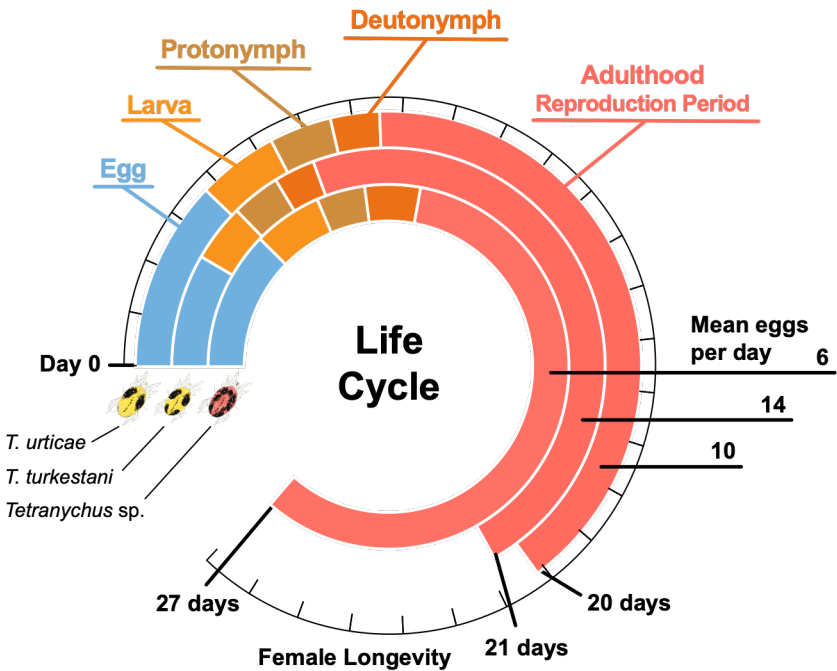
The Spider Mite Team

Department of Biology
Western University

Niagara Falls, ON
September 2026

Life cycle and growth stages of spider mites

Life cycle of *Tetranychus* mites. The three species share the same general developmental stages: egg, larva, protonymph, deutonymph, and adult. The photograph illustrates the life stages of *T. urticae* as an example. Development and the duration of each stage can vary with species and environmental conditions.



Key female characteristics for species identification



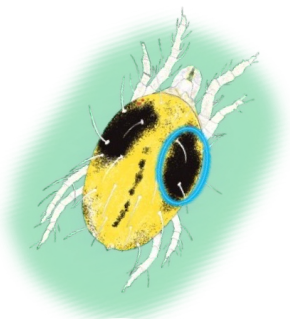
T. urticae
two-spotted spider mite



Tetranychus turkestanii
four-spotted spider mite
not reported in Ontario previously



Tetranychus sp.
red spider mite
not described previously



Morphology of adult females. Adult females exhibit distinct coloration and dorsal spot patterns: *T. urticae* is typically yellowish-green and bears two characteristic dorsal spots; *T. turkestanii* is yellowish-green and typically displays four dorsal spots; The red *Tetranychus* sp. is reddish in coloration and exhibits six dorsal spots extending across most of the dorsum.



Key male taxonomic characteristic for species identification



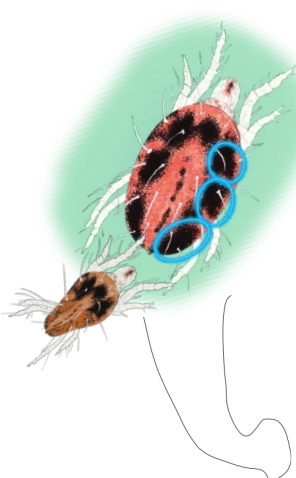
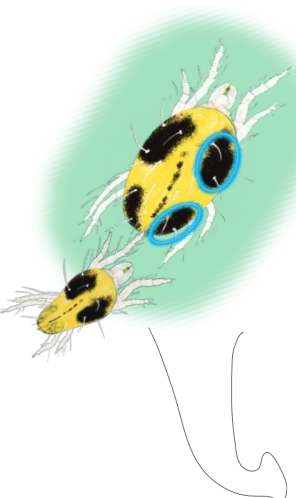
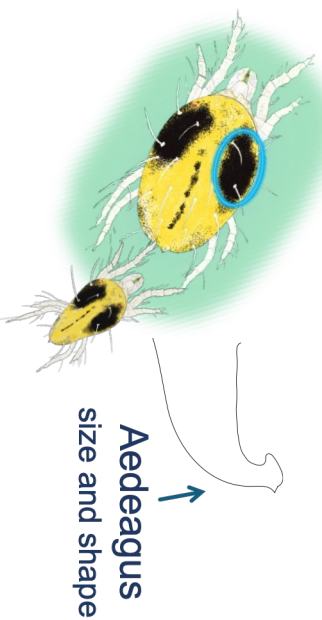
T. urticae
two-spotted spider mite



Tetranychus turkestanii
four-spotted spider mite
not reported in Ontario previously



Tetranychus sp.
red spider mite
not described previously



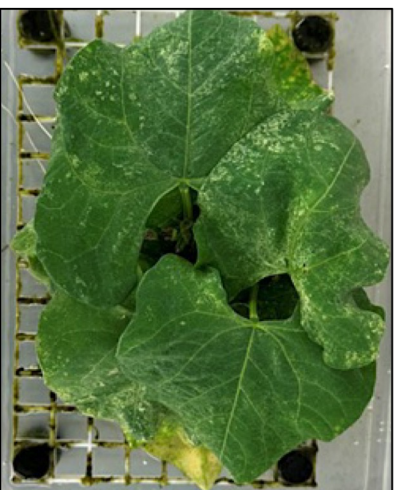
Adult aedeagus (copulatory organ). Male spider mites are generally less distinctive in appearance than females, making species identification based on external features more difficult. The shape and size of the aedeagus provides an important characteristic for distinguishing closely related *Tetranychus* species: *T. urticae* has a small, rounded terminal knob; *T. turkestanii* has a larger, broader knob; and the red *Tetranychus* sp. has a very narrow, elongated aedeagus ending in a distinctive backward-pointing tip. These differences can help confirm species identification under a microscope.



Symptoms and leaf damage caused by a severe mite infestation



T. urticae
two-spotted spider mite
common species



Tetranychus sp.
red spider mite
not previously described



T. turkestanii
four-spotted spider mite
not reported in Ontario



chlorosis

Locally induced



bronzing/necrosis

Systemically induced

Symptoms vary from common **chlorotic stippling** (white spots) to **necrosis** (bronzing). The damage includes loss of photosynthetic tissue, leaf desiccation, premature leaf drop, and diminished canopy function.



Damage Patterns

vs

Mite Species

T. urticae



T. turkestanii



Tetranychus sp.



T. urticae causes widespread whitish chlorotic stippling, which becomes generalized leaf yellowing under severe infestations. In contrast, *T. turkestanii* initially induces only limited chlorotic stippling, but symptoms rapidly develop into generalized leaf necrosis, leading to leaf desiccation and premature leaf drop. The new **red** *Tetranychus* sp. produces intense whitish chlorotic stippling, especially along the leaf veins, with minimal leaf yellowing.



T. urticae Damage

T. urticae causes **widespread whitish chlorotic stippling**, which becomes generalized leaf yellowing under severe infestations.



T. urticae



T. turkestanii



T. turkestanii Damage

T. turkestanii initially induces only limited chlorotic stippling, but symptoms **rapidly develop into generalized leaf necrosis**, leading to leaf desiccation and premature leaf drop.



Red *Tetranychus* sp. Damage

Tetranychus sp.



The new red *Tetranychus* sp. produces **intense whitish aggregated chlorotic stippling**, especially **along the leaf veins**, with minimal leaf yellowing.



Are you having **issues with mites?** Finding **unusual damage?**

Send us samples, we can help you!

For over a decade, the **Grbić Lab** at Western University has partnered with growers and stakeholders to study spider mites in Ontario and promote sustainable agriculture.

Samples shipment to:

Vojislava Grbic
(Attention of Vlad Zhurov)
ScienceStores
Western University
1151 Richmond Street
Chemistry Bldg., Rm.003,
Dock 11
London, ON, N6A5B7
T: 1-519-661-2168

Courier	Account number
Purolator	3973672
FedEx	123803205

Internal reference number:
BAAZ621850

New Spider Mites in Ontario
Scan here for full brochure



Grbić Lab
The Spider Mite Team

